

Generalized EIT No-Go/Go Theory Applicable to Arbitrary Materials

Integrated version: v6.2 — sector-resolved finite-dimensional Markovian weak-probe formulation

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Objective: To determine whether a specified EIT channel in a finite-dimensional Markovian quantum system—including multiple excited levels, internal-state mixing, phonons, dissipation, external fields, polarization, and static inhomogeneity—belongs to the exact sector-resolved no-go, asymptotic-suppression, kernel-go, practical-no-go, or observable-go class in the stationary rotating-frame weak-probe regime.

0. Executive summary

This theory does not classify EIT from the material name alone. The object to be classified is, at minimum,

$$\mathfrak{C} = \{\text{material}, \mathcal{H}_g, \mathcal{H}_e, \text{lower-state pair}, \text{probe/control polarization}, \text{external fields}, T, \text{observable}, \mathbb{S}\}$$

a **material- and channel-resolved configuration** specified by the above tuple. Even within the same material, a spin- Λ channel, an orbital- Λ channel, different polarizations, and different magnetic-field or strain conditions are distinct classification targets.

The theory rigorously separates the following four concepts.

1. Optical dark subspace

Instantaneous optical decoupling is determined by the nullspace of the coupling map Ω .

2. Stationary Lindblad dark state

For a pure dark state to be stationary, additional conditions on the lower-manifold dynamics and on every Lindblad jump operator are required.

3. Coherent EIT pathway

In the reduced model, the excited-manifold resolvent kernel

$$\mathbf{K}(z) = C^\dagger A(z)^{-1} C$$

has off-diagonal elements that determine the feedback between the probe optical sector and the long-lived coherence sector.

4. Observable EIT

This is determined by the response including the full Liouvillian, ground-state coherence, populations, control power, optical depth, inhomogeneous broadening, propagation, and the detection scheme.

In this version, a no-go condition in the full Liouvillian is not defined as a zero of the total susceptibility. We fix a physically specified long-lived sector $\mathbb{S}$ and its complementary direct-response sector $\mathbb{X}$. Let $\chi_{\text{cut}}^{(\mathbb{S})}$ denote the counterfactual response obtained by cutting only the cross-feedback with $\mathbb{S}$, and let χ_{full} denote the complete response. We then define

$$\delta\chi_{\mathbb{S}} \equiv \chi_{\text{full}} - \chi_{\text{cut}}^{(\mathbb{S})}$$

as the **$\mathbb{S}$ -sector-mediated response modification**. When $\mathbb{S}$ is a long-lived off-diagonal coherence sector and the cross-coupling is generated by the control Hamiltonian, this quantity may be interpreted as the coherent EIT correction. A generic Liouvillian block can also contain dissipative coupling, so in that case it should not be called merely a coherent correction.

A structural exact EIT no-go condition is that, for the specified sector partition and parameter domain,

$$\boxed{\delta\chi_{\mathbb{S}}(\theta) \equiv 0}$$

holds identically. This is not an unconditional scalar label intrinsic to the system; it is a sector-resolved statement about the sector $\mathbb{S}$ included in the configuration tuple. By contrast,

$$\chi_{\text{full}} = 0$$

may represent complete transparency, i.e., ideal EIT, and therefore is not a no-go condition.

For a closed two-lower-state weak-probe block, taking $\mathbb{S} = \text{span}\{\rho_{g_2g_1}^{(1)}\}$, one obtains at a regular point

$$\delta\Xi_{\mathbb{S}} = -\frac{\beta K_{12}K_{21}}{\gamma_g + \beta S_2}.$$

In this scalar block, the cross-coupling originates from the control Hamiltonian, so $\delta\Xi_{\mathbb{S}}$ coincides with the coherent EIT correction.

Therefore, for nonzero control and a finite denominator,

$$\boxed{K_{12}K_{21} = 0}$$

is the necessary and sufficient zero condition for the targeted coherence feedback.

Furthermore,

- Under strong full-rank damping, the first nonzero resolvent moment determines the suppression exponent.
- An exact zero results if all moments vanish, or if a generator-preserving reducing symmetry separates the target transfer.
- If the damping operator is singular, an $O(1)$ channel protected in $\ker D$ may remain.
- If the ground-state index is mixed, the full Liouville-space response operator must be used instead of a Hilbert-space kernel.
- Exact $\mathbb{S}$ -sector no-go, asymptotic suppression, practical visibility, and observable EIT must be reported separately.

Accordingly, the present theory does not mean that “all arbitrary materials have already been classified automatically.” Its scope is an **EIT channel that possesses a Markovian GKSL generator on a finite-dimensional Hilbert/Liouville space and admits a weak-probe linear response in a stationary rotating frame**. Infinite-dimensional continua, genuinely non-Markovian memory, strong-probe nonlinear response, and propagation-dominated collective EIT lie outside the scope of the main theorem.

1. Scope and Unit of the Claim

1.1 Classifying channels rather than materials

The no-go/go classification is defined for each configuration tuple below.

$$\mathfrak{C} = (M, G, E, a, b, \epsilon_p, \epsilon_c, \mathbf{F}, T, \mathcal{O}, \mathbb{S}),$$

Here,

- M : material or emitter
- $G \subset \mathcal{H}_g$: lower manifold used in the channel
- $E \subset \mathcal{H}_e$: operative excited manifold
- a, b : lower-state pair
- ϵ_p, ϵ_c : probe/control polarization
- $\mathbf{F}$: magnetic field, electric field, strain, cavity, etc.
- T : temperature or bath condition
- $\mathcal{O}$: observable such as absorption, transmission, or fluorescence

- $\mathbb{S}$: targeted long-lived coherence sector for the no-go/go classification

Thus,

the following are distinct classification targets.

- zero-field ZPL spin- Λ and transverse-field spin- Λ channels in the same NV center
- spin- Λ and orbital- Λ channels in the same group-IV center
- parallel and cross polarizations for the same lower-state pair
- a resolved manifold and a thermally merged manifold in the same material

1.2 Meaning of “applicable to arbitrary materials”

The claim that the theory is material independent means the following.

Given the material-specific Hamiltonian, dipole matrix elements, jump operators, bath rates, and external-field conditions, the same mathematical classifier can evaluate the exact zero, asymptotic order, coherent-pathway permission, and observable visibility of an EIT channel.

It does not mean the following.

- A no-go/go result can be returned from the material name alone.
- $K \neq 0$ guarantees observable EIT.
- Orbital degeneracy necessarily implies a go result.
- The response automatically becomes scalar when the linewidth exceeds the fine-structure splitting.
- The finite-dimensional moment theorem applies without modification to every non-Markovian continuum.

1.3 Assumptions of the main theorems

The exact theorems and solver validation in this framework are restricted to the following assumptions.

1. Finite dimensionality

The Hilbert space, or the adopted Liouville response space, is finite dimensional. When a large vibronic manifold is truncated, convergence of both the observable and the kernel must be checked separately.

2. Markovian GKSL dynamics

The zeroth-order dynamics, including the control field, external fields, and dissipation, is represented in a stationary rotating frame by a time-independent GKSL generator $\mathcal{L}_c$.

3. Weak-probe linear response

The response is retained to $O(\epsilon)$ in the probe amplitude ϵ . The control may be treated nonperturbatively, but probe-induced saturation and population redistribution are not included in the main theorem.

4. Stationary response problem

The system has a unique steady state and a finite gap, or a group/Drazin inverse can be specified uniquely on a compatible trace-zero subspace.

5. Physically fixed sector partition

Before a no-go statement is made, the long-lived sector $\mathbb{S}$, direct-response sector $\mathbb{X}$, source, and detector are physically specified.

6. Analytic-domain claim

If an identically vanishing transfer is claimed over an entire parameter domain, the matrix family and source/detector must be real analytic, or holomorphic after an appropriate complexification.

The following cases are outside the scope of the main theorem.

- Systems with a genuinely non-Markovian memory kernel for which a finite-dimensional Markovian embedding has not converged
- arbitrary strong-probesaturationbistability
- Systems in which propagation, cavity self-consistency, or many-body interactions cannot be separated from the local response
- Time-periodic drives that have not been reduced to a finite-dimensional Floquet–Liouville generator

These cases can be treated by separate extensions, but they are not included in the present claim of applicability to arbitrary materials.

2. General Open-System Formulation

2.1 Hilbert-space decomposition

Decompose the total Hilbert space as

$$\mathcal{H} = \mathcal{H}_e \oplus \mathcal{H}_g \oplus \mathcal{H}_r$$

where $\mathcal{H}_e$ is the selected excited manifold, $\mathcal{H}_g$ is the lower manifold, and $\mathcal{H}_r$ is an auxiliary space containing singlets, charge states, and other loss channels.

Define the optical-coherence block of the density operator by

$$\sigma = P_e \rho P_g \in \mathcal{B}(\mathcal{H}_g, \mathcal{H}_e)$$

.

2.2 Action of excited-confined jumps on optical coherence

When a jump operator is confined to the excited manifold,

$$L_\mu = P_e L_\mu P_e,$$

the Lindblad dissipator acts on the optical-coherence block as

$$P_e \mathcal{D}[L_\mu](\rho) P_g = -\frac{1}{2} L_\mu^\dagger L_\mu \sigma$$

. The recycling term $L_\mu \rho L_\mu^\dagger$ vanishes in the rectangular $e - g$ block.

Thus, even a jump that generates exchange among excited-state populations acts in general as left-loss damping on an optical coherence. This is why the present theory does not use fixed-point projection as a general rule and instead centers the analysis on full-rank damping and inverse-rate expansions.

2.3 Liouville-space generator

The undriven optical-coherence equation can generally be written as

$$\dot{\sigma} = -i(H_e \sigma - \sigma H_g) - \frac{1}{2} R_e \sigma + \mathcal{J}_{eg}[\sigma] + \text{sources},$$

$$R_e = \sum_{\mu} L_\mu^\dagger L_\mu,$$

. Upon vectorization,

$$\text{vec}(AXB) = (B^\top \otimes A) \text{vec } X$$

and hence

$$\mathbb{A}_{eg} = I_g \otimes \left(\frac{1}{2} R_e + i H_e \right) - i H_g^\top \otimes I_e + \mathbb{A}_{\text{corr}}$$

is obtained. Here $\mathbb{A}_{\text{corr}}$ includes ground-state mixing, correlated jumps, spectral-diffusion models, and related corrections.

2.4 Conditions for a Hilbert-space reduction

If the lower state $|g_j\rangle$ is an eigenstate of H_g and the ground-state index j is not mixed in the linear-response optical-coherence block,

$$\sigma = |x_j\rangle\langle g_j|$$

the corresponding subspace is invariant, and the problem reduces exactly to the excited-manifold Hilbert-space resolvent

$$A_j(z)|x_j\rangle = |s_j\rangle$$

.

The full Liouville-space operator must be retained in the following cases.

- Ground-state relaxation mixes the right index.
- Correlated jumps act on both the excited and ground manifolds.
- The control field strongly couples several ground-state coherences.
- Population redistribution cannot be neglected.
- A drive other than the probe is treated nonperturbatively.

3. Optical couplingoptical dark subspacestationary dark state

3.1 Coupling map

Define the lower-to-excited optical coupling in the rotating frame by

$$V_+ = P_e H_{\text{drive}} P_g = \frac{1}{2}\Omega, \quad \Omega : \mathcal{H}_g \rightarrow \mathcal{H}_e$$

. For a lower basis $\{|g_a\rangle\}_{a=1}^{N_g}$,

$$\Omega = (\Omega_1 d_1, \dots, \Omega_{N_g} d_{N_g}),$$

$$(d_a)_j = \langle e_j | \epsilon_a \cdot \hat{\mathbf{d}} | g_a \rangle.$$

The complex phases of the dipole vectors are retained. Under the basis transformation U_e on $\mathcal{H}_e$ and U_g on $\mathcal{H}_g$,

$$\Omega \mapsto U_e \Omega U_g^\dagger$$

so the rank and nullity are basis invariant.

3.2 Theorem 1A — Optical dark-subspace rank

Theorem. A lower-state vector $|v\rangle \in \mathcal{H}_g$ is instantaneously decoupled from optical excitation, namely,

$$V_+ |v\rangle = 0$$

if and only if

$$\Omega |v\rangle = 0$$

. Therefore, the optical dark subspace

$$\mathcal{D}_{\text{opt}} = \ker \Omega$$

has dimension

$$\dim \mathcal{D}_{\text{opt}} = N_g - \text{rank } \Omega$$

.

Proof. Since $V_+ = \Omega/2$, the first equivalence follows directly from the definition. The dimension formula follows from the rank-nullity theorem for finite-dimensional linear maps. $\square$

For a two-lower-state system with both field amplitudes nonzero, a nontrivial optical dark vector exists if and only if

$$\text{rank}(d_1, d_2) < 2.$$

In particular, if both dipole vectors are nonzero,

$$\text{rank}(d_1, d_2) = 1$$

.

3.3 Distinction between an optical dark vector and a Hamiltonian dark eigenstate

Even when $|D\rangle \in \ker \Omega$, it is not a stationary dark state if the lower-manifold Hamiltonian rotates the state from the dark subspace into the bright subspace.

Write the rotating-frame Hamiltonian as

$$H = \begin{pmatrix} H_e & \Omega/2 \\ \Omega^\dagger/2 & H_g^{(\text{rot})} \end{pmatrix}$$

. A pure state $|D\rangle$ fully supported in the lower manifold is a Hamiltonian eigenstate if and only if

$$\Omega|D\rangle = 0, \quad H_g^{(\text{rot})}|D\rangle = E_D|D\rangle$$

.

More generally, the entire dark subspace is invariant under the Hamiltonian if and only if, with P_D denoting the orthogonal projector onto $\ker \Omega$,

$$(I - P_D)H_g^{(\text{rot})}P_D = 0$$

. In the usual two-lower-state Λ system, this condition incorporates two-photon resonance and lower-state energy matching.

3.4 Theorem 1B — Full Lindblad stationary pure dark state

Let the full master equation be

$$\dot{\rho} = \mathcal{L}(\rho) = -i[H, \rho] + \sum_{\mu} \left(L_{\mu} \rho L_{\mu}^{\dagger} - \frac{1}{2} \{L_{\mu}^{\dagger} L_{\mu}, \rho\} \right)$$

.

Theorem. For a normalized vector $|D\rangle$, the pure state $\rho_D = |D\rangle\langle D|$ is stationary, i.e., $\mathcal{L}(\rho_D) = 0$, if and only if there exist complex numbers λ_{μ} and a real number h such that

$$L_\mu |D\rangle = \lambda_\mu |D\rangle \quad (\forall \mu)$$

and

$$\left[H + \frac{i}{2} \sum_\mu (\lambda_\mu^* L_\mu - \lambda_\mu L_\mu^\dagger) \right] |D\rangle = h |D\rangle$$

hold.

Proof sketch. For a stationary pure state, the time derivative of the purity vanishes. For the Lindblad equation,

$$\frac{d}{dt} \text{Tr}(\rho_D^2) = -2 \sum_\mu (\langle L_\mu^\dagger L_\mu \rangle_D - |\langle L_\mu \rangle_D|^2)$$

and each term is a nonnegative variance. Therefore, for every jump operator, $L_\mu |D\rangle = \lambda_\mu |D\rangle$ is necessary. Substituting this condition into the master equation and projecting from the left with $Q_D = I - |D\rangle\langle D|$, one obtains

$$Q_D \left[H + \frac{i}{2} \sum_\mu (\lambda_\mu^* L_\mu - \lambda_\mu L_\mu^\dagger) \right] |D\rangle = 0$$

. The operator in square brackets is Hermitian, so its action is parallel to $|D\rangle$ and has a real eigenvalue h . Conversely, substituting the two conditions into the master equation eliminates all off-diagonal components; trace preservation and Hermiticity then give $\mathcal{L}(\rho_D) = 0$. $\square$

Application. If the EIT dark state is required to lie in the lower manifold, impose additionally

$$P_e |D\rangle = 0, \quad \Omega |D\rangle = 0$$

. When a spontaneous-emission jump starts in an excited state, one normally has $L_\mu |D\rangle = 0$. By contrast, if ground-state dephasing, ground-state relaxation, microwave noise, or another process violates the common-eigenvector condition for $|D\rangle$, an optical dark vector may exist while no stationary pure dark state exists.

3.5 Fixed terminology for classification

This theory distinguishes the following notions.

1. **optical dark capable:** $\dim \ker \Omega > 0$
2. **Hamiltonian dark eigenstate:** $\Omega |D\rangle = 0$ and $H_g^{(\text{rot})} |D\rangle = E_D |D\rangle$
3. **stationary Lindblad dark state:** satisfies Theorem 1B
4. **observable EIT:** a narrow coherent transparency feature appears in the full linear response

Accordingly, neither stationary EIT nor observable EIT is inferred from the rank theorem alone.

4. Basis-invariant coherent-response kernel

4.1 Reduced Hilbert-space kernel

Let the optical-coherence generator be $A(z)$, and define the Green operator by

$$G(z) = A(z)^{-1}$$

. For the multichannel coupling matrix

$$C = (d_1, \dots, d_{N_g})$$

, define

$$\boxed{\mathbf{K}(z) = C^\dagger G(z) C}$$

as the coherent-response matrix kernel.

Its components are

$$K_{ab}(z) = d_a^\dagger G(z) d_b.$$

The elements K_{ab} are invariant under a unitary change of basis in the excited manifold,

$$A \mapsto U A U^\dagger, \quad d_a \mapsto U d_a$$

.

4.2 Physical meaning

- diagonal element

$$S_a = K_{aa} = d_a^\dagger G d_a$$

represents the single-leg optical response or self-energy.

- off-diagonal element

$$K_{ab} = d_a^\dagger G d_b$$

represents the coherent Raman/interference vertex mediated by the excited manifold.

When adiabatic elimination is valid, the effective coherent Hamiltonian in the lower manifold contains, schematically,

$$\langle g_a | H_{\text{eff}} | g_b \rangle \propto -\Omega_a^* \Omega_b K_{ab}$$

. The sign and numerical prefactor depend on the rotating-frame convention.

4.3 Full Liouville-space kernel

When a Hilbert-space reduction is not possible, introduce a source superoperator $\mathbb{B}_b$ and a detection functional $\mathbb{D}_a$, and define

$$\boxed{\mathbb{K}_{ab}(z) = \langle \langle \mathbb{D}_a | \mathbb{A}(z)^{-1} | \mathbb{B}_b \rangle \rangle}$$

. This transfer function is a generic source-to-detector kernel for an arbitrary finite-dimensional Liouvillian. If it is chosen to represent the total probe susceptibility, however, its zero may correspond to complete transparency. The EIT-specific no-go object is the sector-resolved correction $\delta\chi_{\mathbb{S}}$ defined in Section 10.

Moment, Krylov, sector-symmetry, and adjugate tests are applied to the factors of the targeted cross-transfer, or to an augmented finite-dimensional transfer realization representing the full/cut response difference. A zero of the total susceptibility is not called an EIT no-go directly.

4.4 A kernel zero is not equivalent to a dark state

In general,

$$K_{12} = 0 \not\Rightarrow \ker \Omega \neq \{0\}.$$

- Even when the coupling vectors are linearly independent, cancellation among resolvent pathways can yield $K_{12} = 0$.
- Even when an exact dark state exists, dissipation or population dynamics may render the narrow EIT feature unobservable.

5. Fast dampingrate mappingmoment hierarchy

5.1 Standard decomposition

Decompose the generator on the selected finite-dimensional response space as

$$A_\Gamma(z) = \Gamma D + A_0(z)$$

.

- $\Gamma > 0$: selected fast-rate scale
- D : dimensionless fast-generator shape
- $A_0(z)$: detuning, fine structure, slow loss, fields, strain, and related terms

Important: Γ is not in general identical to a population-relaxation rate quoted in the literature. The term ΓD must be defined from the matrix that actually appears in the optical-coherence equation.

5.2 Microscopic mapping from population hopping to optical-coherence damping

Define incoherent hopping between the excited branches $|X\rangle$ and $|Y\rangle$ by

$$L_{Y \leftarrow X} = \sqrt{k_{Y \leftarrow X}} |Y\rangle \langle X|, \quad L_{X \leftarrow Y} = \sqrt{k_{X \leftarrow Y}} |X\rangle \langle Y|$$

.

The population equations are

$$\dot{p}_X = -k_{Y \leftarrow X} p_X + k_{X \leftarrow Y} p_Y,$$

$$\dot{p}_Y = k_{Y \leftarrow X} p_X - k_{X \leftarrow Y} p_Y.$$

The population mode measured relative to equilibrium relaxes at the rate

$$\Gamma_{\text{pop}} = k_{Y \leftarrow X} + k_{X \leftarrow Y}$$

.

By contrast, for the optical coherences associated with an arbitrary lower state $|g\rangle$,

$$\sigma_{Xg} = \langle X | \rho | g \rangle, \quad \sigma_{Yg} = \langle Y | \rho | g \rangle$$

the recycling term does not contribute, and

$$\boxed{\dot{\sigma}_{Xg}|_{\text{hop}} = -\frac{k_{Y \leftarrow X}}{2} \sigma_{Xg}}$$

$$\dot{\sigma}_{Yg}|_{\text{hop}} = -\frac{k_{X \leftarrow Y}}{2} \sigma_{Yg}$$

.

Proof. For example, for $L_{Y \leftarrow X}$,

$$P_e L \rho L^\dagger P_g = 0, \quad L^\dagger L = k_{Y \leftarrow X} |X\rangle\langle X|,$$

so only the left action of the anticommutator remains in the $X - g$ block, with coefficient $-k_{Y \leftarrow X}/2$. $\square$

Symmetric two-branch convention

$$k_{Y \leftarrow X} = k_{X \leftarrow Y} = k$$

and suppose the literature or model defines

$$(p_X - p_Y) = -\Gamma_{XY}(p_X - p_Y)$$

. Then

$$\Gamma_{XY} = 2k$$

. Consequently, the additional damping of each optical coherence is

$$\gamma_{\text{oc}}^{(XY)} = \frac{k}{2} = \frac{\Gamma_{XY}}{4}$$

.

Thus, if $D = I$ is chosen on two equivalent branches, the scale used in the moment expansion must be

$$\Gamma = \gamma_{\text{oc}}^{(XY)} = \Gamma_{XY}/4$$

. Alternatively, if one sets $\Gamma = \Gamma_{XY}$, then one must use $D = I/4$. These two conventions must not be mixed.

Asymmetric or detailed-balance hopping

If $k_{Y \leftarrow X} \neq k_{X \leftarrow Y}$,

$$A_{\text{hop}} = \frac{1}{2} \begin{pmatrix} k_{Y \leftarrow X} & 0 \\ 0 & k_{X \leftarrow Y} \end{pmatrix}$$

and in general this is not $\Gamma_{\text{pop}}/4$ times the identity. Material-specific calculations must input the upward and downward rates separately rather than replacing them by an inaccurate scalar rate.

More generally, define the total incoherent escape rate from an excited state $|e_j\rangle$ by

$$r_j^{\text{out}} = \sum_{m \neq j} k_{m \leftarrow j} + \sum_{r \notin E} k_{r \leftarrow j}$$

. The corresponding optical coherence then acquires the damping contribution

$$\frac{1}{2} r_j^{\text{out}}$$

. Pure dephasing must be derived separately from the adopted jump-operator normalization.

5.3 Algebraic resolvent reduction

For the exact algebraic result in this section, assume initially only that D is invertible. With the definitions

$$X(z) = -D^{-1}A_0(z), \quad \nu = D^{-1}c$$

,

$$A_\Gamma = D(\Gamma I - X)$$

so, at every point where A_Γ is invertible,

$$\boxed{K_\Gamma(z) = p^\dagger A_\Gamma(z)^{-1}c = p^\dagger(\Gamma I - X(z))^{-1}\nu}$$

.

5.4 Controlled large-rate expansion

Under the condition

$$q(z) = \frac{\|X(z)\|}{\Gamma} = \frac{\|D^{-1}A_0(z)\|}{\Gamma} < 1$$

,

$$(\Gamma I - X)^{-1} = \frac{1}{\Gamma} \sum_{n=0}^{\infty} \frac{X^n}{\Gamma^n}$$

. Therefore,

$$\boxed{K_\Gamma(z) = \sum_{n=0}^{\infty} \frac{M_n(z)}{\Gamma^{n+1}}}$$

$$\boxed{M_n(z) = p^\dagger X(z)^n \nu, \quad \nu = D^{-1}c}$$

. The former notation mixed u and ν ; this version consistently uses ν .

The remainder after truncation at order m ,

$$R_m = K_\Gamma - \sum_{n=0}^m \frac{M_n}{\Gamma^{n+1}}$$

, satisfies

$$\boxed{|R_m| \leq \frac{\|p\| \|D^{-1}\| \|c\| q^{m+1}}{\Gamma} \frac{1}{1-q}}$$

.

5.5 Theorem 3 — First nonzero moment

$$M_0 = \cdots = M_{m-1} = 0, \quad M_m \neq 0$$

then, as $\Gamma \rightarrow \infty$,

$$K_\Gamma = \frac{M_m}{\Gamma^{m+1}} + O(\Gamma^{-m-2})$$

. This follows directly from the Neumann expansion.

5.6 Theorem 4 — Exact finite-dimensional transfer zero: complete equivalence

Let the response-space dimension be N , and assume that D is invertible at fixed z . The following statements are equivalent.

1. $M_n = p^\dagger X^n \nu = 0$ for $n = 0, \dots, N-1$
2. p is orthogonal to the finite Krylov space

$$\mathcal{K}_N(X, \nu) = \text{span}\{\nu, X\nu, \dots, X^{N-1}\nu\}$$

3. For every polynomial q ,

$$p^\dagger q(X)\nu = 0$$

4. For every $\Gamma \notin \text{spec } X$,

$$K_\Gamma = 0$$

Proof. Statements 1 and 2 are equivalent by definition. By the Cayley–Hamilton theorem, X^N is a linear combination of $I, X, \dots, X^{N-1}$. Hence statement 1 implies inductively that every M_n vanishes, which yields statement 3.

Next, for $\Gamma \notin \text{spec } X$,

$$(\Gamma I - X)^{-1} = \frac{\text{adj}(\Gamma I - X)}{\det(\Gamma I - X)}.$$

The adjugate $\text{adj}(\Gamma I - X)$ can be expressed as a polynomial in X of degree at most $N-1$. Statement 3 therefore implies

$$p^\dagger (\Gamma I - X)^{-1} \nu = 0$$

which gives statement 4.

Conversely, if statement 4 holds on an open set of sufficiently large Γ , every coefficient in the Laurent expansion in $1/\Gamma$ vanishes. Thus all $M_n = 0$, and in particular statement 1 holds. $\square$

Important: This exact theorem is not restricted to the convergence domain of the Neumann series. The Cayley–Hamilton/adjugate representation extends it to the entire region in which A_Γ is invertible.

To claim an exact transfer zero over an entire domain in z , the equivalent conditions above must hold at every z . Numerical zeros at finitely many frequencies do not prove an identically vanishing transfer.

5.7 Sector graph and suppression order

Decompose the coherence space into orthogonal sectors,

$$\mathcal{V} = \bigoplus_a \mathcal{V}_a$$

and suppose that D and A_{diag} preserve each sector, with

$$A_0 = A_{\text{diag}} + V$$

. Regard one action of V as one edge in the sector graph, and let d be the minimum number of V -insertions required to connect the source sector to the readout sector. Then

$$M_n = 0 \quad (n < d),$$

$$\boxed{K_\Gamma = O(\Gamma^{-1-d})}$$

. If several paths undergo phase cancellation, the actual order can be still higher.

5.8 Finite-rate bound

$$B_\Gamma = \Gamma D + A_{\text{diag}}, \quad A_\Gamma = B_\Gamma + V$$

and assume $p^\dagger B_\Gamma^{-1} c = 0$. The resolvent identity gives

$$K_\Gamma = -p^\dagger B_\Gamma^{-1} V A_\Gamma^{-1} c.$$

$$r_d = \frac{\|D^{-1} A_{\text{diag}}\|}{\Gamma} < 1, \quad r = \frac{\|D^{-1} (A_{\text{diag}} + V)\|}{\Gamma} < 1$$

then

$$\boxed{|K_\Gamma| \leq \frac{\|p\| \|c\| \|D^{-1}\|^2 \|V\|}{\Gamma^2 (1 - r_d)(1 - r)}}$$

.

5.9 Precise scope of the condition $D > 0$

The earlier statement that “the moment theorem requires $D > 0$ ” is corrected as follows.

(A) Exact algebraic transfer zero

Theorem 4 requires only that

$$\boxed{D \text{ is invertible}}$$

. Hermitian positivity is unnecessary.

(B) Neumann asymptotic expansion

The required conditions are

$$D \text{ is invertible}, \quad \|D^{-1} A_0\|/\Gamma < 1.$$

. Positivity is unnecessary.

(C) Physical interpretation as fast damping and uniform stability

A clean sufficient condition ensuring that large Γ damps every mode is

$$\boxed{\text{Re } D = \frac{D + D^\dagger}{2} \geq d_{\min} I > 0}$$

. In particular, $D = D^\dagger > 0$ is sufficient. In this case, $\Gamma d_{\min}$ provides a coercive damping scale.

(D) Hermitian metric overlap

$$H_A = (A_\Gamma + A_\Gamma^\dagger)/2 > 0$$

The bounded geometry proxy η_H defined using

(E) Singular-damping protected-channel theorem

For the present Schur-complement theorem using the orthogonal projector $P = \text{Proj}(\ker D)$, assume

$$D = D^\dagger \geq 0$$

. A non-normal singular D requires a Riesz projector and a generalized nullspace, and is outside the scope of the present main theorem.

(F) Nonnormal fast generator

Even when D is invertible, a non-accretive D can produce a large finite- Γ response through pseudospectral amplification. Algebraic scaling may still be used, but monotonic damping suppression or a simple visibility bound must not be claimed without checking the relevant resolvent norm.

6. Exact symmetry transfer zero on the full generator

6.1 Reduced-space projector theorem

Suppose the response space on which $A(z)$ acts admits orthogonal projectors

$$P_\alpha P_\beta = \delta_{\alpha\beta} P_\alpha, \quad \sum_\alpha P_\alpha = I$$

such that

$$[A(z), P_\alpha] = 0 \quad (\forall \alpha, z)$$

. If the source and readout lie in different sectors,

$$P_\beta c = c, \quad p^\dagger P_\alpha = p^\dagger, \quad \alpha \neq \beta$$

then, at every point where $A(z)$ is invertible,

$$p^\dagger A(z)^{-1} c = 0$$

.

Proof. If $[A, P_\alpha] = 0$, then $[A^{-1}, P_\alpha] = 0$. Therefore,

$$p^\dagger A^{-1} c = p^\dagger P_\alpha A^{-1} P_\beta c = p^\dagger A^{-1} P_\alpha P_\beta c = 0.$$

□

The earlier formulation in terms of a Hermitian symmetry operator $Q = Q^\dagger$ is a special case of this theorem, obtained from its spectral projectors P_α .

6.2 Full-Liouvillian transfer-zero theorem

Let $\mathcal{L}_c$ be the zeroth-order Liouvillian including the control but excluding the probe. Let ρ_0 be the unique steady state and $\langle\langle \mathcal{K} |$ the trace functional, and define

$$\Pi_0 = |\rho_0\rangle\rangle\langle\langle \mathcal{K}|, \quad Q_0 = I - \Pi_0$$

.

Suppose the Liouville space admits orthogonal sector projectors $\mathbb{P}_\alpha$ satisfying

$$[\mathcal{L}_c, \mathbb{P}_\alpha] = 0, \quad [Q_0, \mathbb{P}_\alpha] = 0$$

. Define the probe source and detector by

$$|b_p\rangle\rangle = \mathcal{V}_p|\rho_0\rangle\rangle, \quad \langle\langle d_p|$$

and assume

$$\mathbb{P}_\beta|b_p\rangle\rangle = |b_p\rangle\rangle, \quad \langle\langle d_p|\mathbb{P}_\alpha = \langle\langle d_p|, \quad \alpha \neq \beta$$

. Then the source-to-detector transfer defined through the group/Drazin inverse,

$$\chi_p = -\langle\langle d_p|\mathcal{L}_c^\#|b_p\rangle\rangle$$

vanishes exactly.

Proof. Since $\mathcal{L}_c$, Q_0 , and $\mathbb{P}_\alpha$ commute, the inverse on Q_0 , and hence the group inverse $\mathcal{L}_c^\#$, preserves every sector. The same orthogonality argument as in the reduced-space theorem makes the transfer vanish. $\square$

Interpretive caution. This theorem directly proves a zero of the specified source-to-detector transfer. If it is applied to the total probe susceptibility and yields $\chi_p = 0$, that result may represent complete transparency and therefore is not, by itself, an EIT no-go. To obtain an EIT no-go statement, apply the theorem to the targeted sector correction $\delta\chi_s$ defined in Section 10 or to a cross-sector transfer required to construct it.

6.3 Microscopic sufficient condition for Lindblad symmetry

Let U be a unitary operator on the Hilbert space, and define

$$U(\rho) = U\rho U^\dagger$$

.

$$[\mathcal{L}_c, U] = 0$$

The following conditions are sufficient for

1. The full Hamiltonian, including the control, satisfies

$$UH_cU^\dagger = H_c$$

2. The jump operators are closed under unitary mixing:

$$UL_\mu U^\dagger = \sum_\nu V_{\mu\nu} L_\nu,$$

where V is unitary on the jump-index space, and the Kossakowski matrix/rates are invariant under this transformation.

3. If the steady state is unique, then automatically

$$U\rho_0U^\dagger = \rho_0.$$

. If the steady state is nonunique, choose a symmetry-invariant zeroth-order state.

These conditions include not only Hamiltonian symmetry but also radiative decay, intersystem crossing, orbital hopping, ground-state relaxation, dephasing, and correlated jumps.

6.4 Full-generator symmetry audit

Before declaring an exact symmetry-protected transfer zero, inspect all of the following items.

block	required test
lower Hamiltonian	Does it preserve the symmetry sector?
excited Hamiltonian	Does it preserve the sector after including spin-orbit, spin-spin, Zeeman, and strain terms?
control Hamiltonian	Does it preserve the sector for the selected polarization and phase?
excited-state jumps	Are the jump set and rates symmetry covariant?
radiative/ISC loss	Does it avoid mixing the sectors?
ground-state dephasing/relaxation	Does it avoid mixing the dark/source sector?
zeroth-order steady state	Is it symmetry invariant?
probe source	Does it belong to one well-defined sector?
detector	Does it belong to an orthogonal sector different from the source?

If any one of these conditions fails, an exact symmetry-protected transfer zero does not follow. The symmetry-breaking term should then be treated as V , and the moment hierarchy or perturbative opening should be evaluated.

6.5 Symmetry-protected zeroKrylov zeroaccidental zero

- **Symmetry-protected transfer zero:** enforced by reducing projectors of the full generator and robust against symmetry-preserving perturbations.
- **Krylov exact transfer zero:** every finite transfer path is orthogonal to the readout. This can occur even without an explicit symmetry.
- **Accidental parameter zero:** pathway cancellation occurs only at a particular detuning or field value; it is not a domain-wide zero.

These are called EIT no-go results only when the vanishing transfer is the targeted sector correction of Section 10 or a cross-channel required for that correction. They must be reported separately from a zero of the total susceptibility.

7. Singular Damping and Protected Channels

The orthogonal protected-subspace theorem in this section assumes

$$D = D^\dagger \geq 0$$

. If D is singular, define

$$P = \text{Proj}(\ker D), \quad Q_D = I - P$$

. On the fast subspace, define

$$A_{QQ} = \Gamma Q_D D Q_D + Q_D A_0 Q_D$$

and introduce the Schur complement for the slow/protected sector,

$$S_P(\Gamma) = PA_0P - PA_0Q_DA_{QQ}^{-1}Q_DA_0P$$

.

If PA_0P is invertible on the spectral domain of interest,

$$S_P(\Gamma) = PA_0P + O(\Gamma^{-1})$$

and

$$\boxed{K_\Gamma = p^\dagger P(PA_0P)^{-1}Pc + O(\Gamma^{-1})}$$

.

Consequently, an $O(1)$ coherent channel can survive strong damping only if both the probe and control vectors have projections onto the genuine kernel of the optical-coherence damping operator.

8. Extension to Arbitrary Materials with Multiple Dissipation Rates

8.1 Fixed-ratio fast-rate limit

For a generic material,

$$A = A_0 + \sum_{r=1}^R \Gamma_r D_r$$

. If all fast rates grow along a common scale Γ ,

$$\Gamma_r = \Gamma \lambda_r$$

with fixed λ_r , define

$$D(\boldsymbol{\lambda}) = \sum_r \lambda_r D_r$$

. If $D(\boldsymbol{\lambda})$ is invertible on a given ray, the algebraic moment hierarchy can be applied. If, in addition,

$$\text{Re } D(\boldsymbol{\lambda}) > 0$$

, that ray can be interpreted as a uniform fast-damping limit.

If the suppression exponent depends on $\boldsymbol{\lambda}$, the no-go/go classification should be displayed as a phase diagram in bath-rate space.

8.2 Independent-rate asymptotics

If the Γ_r vary independently, the leading term can depend on the path taken toward the asymptotic limit. One should therefore evaluate the result separately for

- dominant-rate sector
- fixed-ratio rays
- experimentally relevant trajectory $\Gamma_r(T)$

rather than assigning a single “universal exponent.”

8.3 General finite-dimensional transfer-zero criterion and singular points

Let θ be a parameter vector and consider a finite-dimensional matrix family $A(\theta)$. Assume one of the following.

1. θ is complex and $A(\theta)$ is holomorphic on a connected complex domain $\mathcal{U}$.
2. θ is real and $A(\theta)$ is real analytic on a connected open real domain $\mathcal{U}$.

The identity theorem is not invoked for merely smooth parameter dependence.

regular set

$$\mathcal{U}_{\text{reg}} = \{\theta \in \mathcal{U} : \det A(\theta) \neq 0\}$$

one has

$$K(\theta) = p^\dagger A(\theta)^{-1} c = \frac{N(\theta)}{\Delta(\theta)},$$

$$N(\theta) = p^\dagger \text{adj } A(\theta) c, \quad \Delta(\theta) = \det A(\theta).$$

Theorem 7A — Regular-domain exact transfer zero

On a connected analytic domain $\mathcal{U}$, if

$$\boxed{N(\theta) \equiv 0}$$

then, for every $\theta \in \mathcal{U}_{\text{reg}}$,

$$\boxed{K(\theta) = 0}$$

. Conversely, if K vanishes on a nonempty open subset of $\mathcal{U}_{\text{reg}}$, then $N = 0$ on that subset, and the holomorphic or real-analytic identity theorem gives $N \equiv 0$.

Caution: The condition $\text{adj } A = 0$ at a singular point does not by itself imply that the response is finite or zero. If the rank deficiency is at least two, the adjugate can vanish automatically.

8.3.1 Isolated singularity and Laurent test

Suppose θ_0 is an isolated singular point and, along a one-dimensional regularization parameter s ,

$$A(s)^{-1} = \sum_{j=-\nu}^{\infty} R_j s^j$$

admits the Laurent expansion above. The transfer is

$$K(s) = \sum_{j=-\nu}^{\infty} (p^\dagger R_j c) s^j.$$

- The necessary and sufficient condition for pole cancellation is

$$p^\dagger R_j c = 0 \quad (j = -\nu, \dots, -1)$$

- Laurent finite part

$$\boxed{\text{FP}_{s=0} K(s) = p^\dagger R_0 c}$$

- If every negative-power coefficient vanishes, the scalar-transfer singularity is removable, and the analytic continuation is defined by

$$\boxed{K_{\text{cont}}(0) = p^\dagger R_0 c}$$

.

- The continued transfer vanishes at the singular point if all pole coefficients vanish and

$$p^\dagger R_0 c = 0$$

Because the original matrix inverse $A(0)^{-1}$ does not exist, one must not write $K(0) = p^\dagger R_0 c$ without explicitly specifying the regularization.

8.3.2 Semisimple zero and Drazin/group inverse

Suppose the zero eigenvalue of A is semisimple, and let P_0 be the spectral projector onto its nullspace. Under a shift regularization,

$$(A + sI)^{-1} = \frac{P_0}{s} + A^D + O(s),$$

where A^D is the Drazin inverse. Hence

$$K(s) = \frac{p^\dagger P_0 c}{s} + p^\dagger A^D c + O(s).$$

Therefore,

$$\boxed{p^\dagger P_0 c = 0}$$

is the pole-cancellation condition, and

$$\boxed{\text{FP}_{s=0} K(s) = p^\dagger A^D c}$$

- When the pole cancels, this finite part is also the value of the removable continuation.

For the unique steady-state zero mode of a full Liouvillian, the probe source is trace zero and therefore normally has no component along the steady-state projector. The linear response is then defined by the group inverse on the trace-zero response space.

8.3.3 Operational rule

1. At regular points, use the adjugate-numerator identity.
2. For a Liouvillian zero mode, use the trace-zero subspace or group inverse.
3. At an exceptional point or nonsemisimple singularity, inspect every Laurent pole coefficient.
4. Distinguish the finite part, removable continuation, and ordinary inverse.
5. Do not treat $0/0$ as a numerical zero.

9. Weak-probe susceptibility: full Liouvillian derivation and reduced kernel

9.1 Full linear-response formula

Let the probe amplitude be a small complex parameter ϵ , and write

$$\mathcal{L}(\epsilon) = \mathcal{L}_c + \epsilon \mathcal{V}_p + \epsilon^* \tilde{\mathcal{V}}_p + O(|\epsilon|^2)$$

- Here $\mathcal{L}_c$ contains the control, all dissipation, and the external fields. Let the zeroth-order steady state satisfy

$$\mathcal{L}_c \rho^{(0)} = 0, \quad \text{Tr } \rho^{(0)} = 1$$

and expand

$$\rho = \rho^{(0)} + \epsilon \rho^{(1)} + \epsilon^* \tilde{\rho}^{(1)} + O(|\epsilon|^2)$$

. The coefficient of ϵ satisfies

$$\mathcal{L}_c \rho^{(1)} = -\mathcal{V}_p \rho^{(0)}, \quad \text{Tr } \rho^{(1)} = 0$$

.

Assuming a unique stationary state and a finite Liouvillian gap, define the group inverse $\mathcal{L}_c^\#$ on the trace-zero subspace. Then

$$\boxed{\rho^{(1)} = -\mathcal{L}_c^\# \mathcal{V}_p \rho^{(0)}}$$

. Apart from the normalization constant χ_0 , the linear susceptibility of an observable O_p is

$$\boxed{\chi_p = -\chi_0 \langle\langle O_p | \mathcal{L}_c^\# | \mathcal{V}_p \rho^{(0)} \rangle\rangle}$$

. This is the reference formula for the full-Liouvillian susceptibility; the zero eigenvalue of the Liouvillian is not handled with an ordinary inverse.

9.2 Closed two-lower-state first-order block

Among the first-order variables, let $x \in \mathcal{H}_e$ be the probe optical-coherence vector and let the lower-state coherence be

$$s = \rho_{g_2 g_1}^{(1)}$$

. Assume that the remaining first-order variables either decouple or have been eliminated exactly by a Schur complement.

Fix the sign convention as

$$\begin{pmatrix} A & i\Omega_c d_2/2 \\ i\Omega_c^* d_2^\dagger/2 & \gamma_g \end{pmatrix} \begin{pmatrix} x \\ s \end{pmatrix} = \begin{pmatrix} i\Omega_p d_1/2 \\ 0 \end{pmatrix}$$

. Here,

- A : effective optical-coherence generator excluding the control; it may include self-energies from other blocks when necessary.
- $\gamma_g = \gamma_{12} - i\delta_2$: lower-coherence complex generator
- d_1, d_2 : probe/control dipole vectors

Assume A is invertible and set $G = A^{-1}$. The first row gives

$$x = G \frac{i\Omega_p}{2} d_1 - G \frac{i\Omega_c}{2} d_2 s.$$

Substituting this into the second row, without dividing by γ_g itself, gives

$$(\gamma_g + \beta S_2) s = -\frac{i\Omega_c^*}{2} d_2^\dagger G \frac{i\Omega_p}{2} d_1, \quad \beta = \frac{|\Omega_c|^2}{4},$$

$$S_2 = d_2^\dagger G d_2$$

. Therefore, the ideal limit $\gamma_g = 0$ is covered by the same expression provided that the combined denominator $\gamma_g + \beta S_2 \neq 0$.

9.3 Theorem 2A — Reduced susceptibility as an exact block Schur complement

$$S_i = d_i^\dagger G d_i, \quad K_{ij} = d_i^\dagger G d_j$$

Define the normalized probe response by

$$\Xi = \frac{d_1^\dagger x}{i\Omega_p/2}$$

. Then

$$\Xi = S_1 - \frac{\beta K_{12} K_{21}}{\gamma_g + \beta S_2}$$

.

Proof. The lower-coherence equation above gives

$$s = \frac{\Omega_c^* \Omega_p}{4} \frac{K_{21}}{\gamma_g + \beta S_2}$$

. Substituting this into

$$x = G \frac{i\Omega_p}{2} d_1 - G \frac{i\Omega_c}{2} d_2 s$$

, acting from the left with $d_1^\dagger$, and dividing by $i\Omega_p/2$ yields the result. This is the Schur complement of the first-order Liouvillian block and requires no additional adiabatic approximation. $\square$

The control-induced coherent correction is

$$\Delta\Xi = \frac{\beta K_{12} K_{21}}{\gamma_g + \beta S_2}$$

. At a regular point with nonzero control $\beta \neq 0$ and a finite denominator,

$$K_{12} K_{21} = 0$$

is the necessary and sufficient zero condition for the complex coherent correction $\Delta\Xi$. If the denominator vanishes, use the Laurent/Drazin analysis of Section 8.3.

9.4 Conditions for agreement with the full Liouvillian

The reduced expression above agrees with the full formula under the following conditions.

1. The response is first order in the probe.
2. The zeroth-order state $\rho^{(0)}$, including the control, has been solved correctly.
3. The first-order source reduces to the selected probe vector d_1 .
4. First-order variables other than (x, s) either decouple or are absorbed into A, γ_g, d_i through a Schur complement.
5. Population saturation and probe-induced redistribution are $O(|\Omega_p|^2)$ and therefore do not enter at first order.
6. The detector corresponds to $d_1^\dagger x$.

If these conditions fail, promote the scalar s to a ground-coherence vector and use the full block-matrix Schur complement or $\mathcal{L}_c^\#$ directly.

9.5 General matrix ground-coherence version

For several ground-state coherences $s \in \mathbb{C}^m$, write the first-order block as

$$\begin{pmatrix} A & B \\ C & G_g \end{pmatrix} \begin{pmatrix} x \\ s \end{pmatrix} = \begin{pmatrix} b_p \\ 0 \end{pmatrix}$$

. Here the source and detector are assumed to belong to the optical-response sector x .

If A is invertible, the first row gives

$$x = A^{-1}b_p - A^{-1}Bs$$

and, using the ground-coherence Schur complement

$$S_g = G_g - CA^{-1}B$$

, one obtains

$$S_g s = -CA^{-1}b_p$$

. If S_g is invertible,

$$s = -S_g^{-1}CA^{-1}b_p$$

and

$$x = A^{-1}b_p + A^{-1}BS_g^{-1}CA^{-1}b_p$$

. This form does not require G_g itself to be invertible and therefore includes the ideal ground-coherence limit.

Only when G_g is also invertible may one use the equivalent alternative Schur complement

$$x = (A - BG_g^{-1}C)^{-1}b_p$$

.

For an optical detector d ,

$$\chi_{\text{full}} = \chi_0 d^\dagger x, \quad \chi_{\text{cut}}^{(\mathbb{S})} = \chi_0 d^\dagger A^{-1}b_p,$$

and therefore the response modification mediated by the specified sector $\mathbb{S}$ is

$$\delta\chi_{\mathbb{S}} = \chi_{\text{full}} - \chi_{\text{cut}}^{(\mathbb{S})} = \chi_0 d^\dagger A^{-1}BS_g^{-1}CA^{-1}b_p$$

. The quantity $\chi_{\text{cut}}^{(\mathbb{S})}$ is not the control-off response; it is an algebraic counterfactual obtained by retaining the same diagonal blocks, source, detector, and zeroth-order state while cutting only the targeted $x \leftrightarrow s$ feedback. The cut generator need not define an independent CPTP dynamics. Once the sector $\mathbb{S}$ is fixed, we abbreviate it as χ_{cut} .

If the source or detector also has a direct component in the ground-coherence sector, define the counterfactual difference using the full block inverse rather than the expression above. For singular A or S_g , use the Laurent/Drazin prescription of Section 8.3.

9.6 Recovery of the textbook three-level Λ model

Consider the standard Λ system with one excited state $|e\rangle$ and two lower states $|g_1\rangle, |g_2\rangle$. Choose the dipole normalization

$$d_1 = d_2 = 1$$

and set

$$A = \gamma_{31} - i\Delta_p, \quad \gamma_g = \gamma_{21} - i\delta_2, \quad \beta = \frac{|\Omega_c|^2}{4}$$

. Then

$$S_1 = S_2 = K_{12} = K_{21} = A^{-1}.$$

Substitution into Theorem 2A gives

$$\begin{aligned} \Xi &= \frac{1}{A} - \frac{\beta/A^2}{\gamma_g + \beta/A} \\ &= \boxed{\frac{\gamma_g}{A\gamma_g + \beta}}. \end{aligned}$$

Therefore, the probe optical coherence is

$$\boxed{\rho_{eg_1}^{(1)} = \frac{i\Omega_p}{2} \frac{\gamma_{21} - i\delta_2}{(\gamma_{31} - i\Delta_p)(\gamma_{21} - i\delta_2) + |\Omega_c|^2/4}}$$

. This is the standard three-level EIT susceptibility in the adopted rotating-frame sign convention.

- control off ($\beta = 0$): $\Xi = A^{-1}$
- exact two-photon resonance and ideal ground coherence ($\delta_2 = 0, \gamma_{21} \rightarrow 0$): $\Xi \rightarrow 0$
- in the same limit, $\delta\Xi_{\text{S}} = \Xi_{\text{full}} - \Xi_{\text{cut}} = -A^{-1} \neq 0$

Thus, a zero of the total susceptibility represents perfect EIT and differs from a zero of the sector-mediated correction. This recovery shows that the framework contains the textbook Λ response as a special case.

9.7 Recovery of a Mishina-type multi-excited-state model

Consider an operative excited manifold consisting of N_e resolved branches, with an optical-coherence generator diagonal in the selected basis. Let λ label an inhomogeneous class, velocity class, or static detuning class, and write

$$A_\lambda = \text{diag}(a_1(\lambda), \dots, a_{N_e}(\lambda)), \quad a_j(\lambda) = \gamma_j - i\Delta_j(\lambda)$$

. For the probe and control dipole vectors

$$d_1 = (d_{11}, \dots, d_{1N_e})^\top, \quad d_2 = (d_{21}, \dots, d_{2N_e})^\top$$

, one obtains

$$\boxed{S_{1,\lambda} = \sum_{j=1}^{N_e} \frac{|d_{1j}|^2}{a_j(\lambda)}}, \quad \boxed{S_{2,\lambda} = \sum_{j=1}^{N_e} \frac{|d_{2j}|^2}{a_j(\lambda)}}$$

and

$$\boxed{K_{12,\lambda} = \sum_{j=1}^{N_e} \frac{d_{1j}^* d_{2j}}{a_j(\lambda)}, \quad K_{21,\lambda} = \sum_{j=1}^{N_e} \frac{d_{2j}^* d_{1j}}{a_j(\lambda)}}$$

. Hence the response of each class is

$$\boxed{\Xi_\lambda = S_{1,\lambda} - \frac{\beta K_{12,\lambda} K_{21,\lambda}}{\gamma_g + \beta S_{2,\lambda}}}.$$

This is the diagonal-resolvent structure of a Mishina-type multilevel susceptibility, containing the coherent sum of Raman amplitudes through multiple excited levels and the self-energy of each optical leg. Static Doppler or inhomogeneous broadening is incorporated only after the local response has been calculated, by averaging

$$\boxed{\bar{\chi} = \int d\lambda P(\lambda) \chi_\lambda}$$

. When optical pumping is included, the zeroth-order populations enter $\rho^{(0)}$ and the source b_p , so they must first be solved with the full Liouvillian. Thus, the present theory contains the **multi-excited-state Green-function interference structure** of Mishina-type theories; it does not unconditionally identify its approximations with the propagation, velocity-changing collisions, or pumping approximations of any particular experiment.

9.8 Observable dictionary

object	definition	role
K_{12}, K_{21}	off-diagonal resolvent	coherent Raman/interference vertex
S_1, S_2	diagonal resolvent	optical response/self-energy
χ	$\chi_0 \Xi$	local linear susceptibility
C_{abs}	normalized absorption contrast	specified observable
T	$\exp[-L\alpha]$	propagated transmission

For a non-normal generator, in general,

$$K_{21} \neq K_{12}^*.$$

so the two quantities must be computed independently.

Hermitian part

$$H_A = (A + A^\dagger)/2 > 0$$

Only when

$$\eta_H = \frac{|d_1^\dagger H_A^{-1} d_2|^2}{(d_1^\dagger H_A^{-1} d_1)(d_2^\dagger H_A^{-1} d_2)}$$

does the bounded geometry proxy satisfy $0 \leq \eta_H \leq 1$.

9.9 Relative absorption contrast and visibility branches

As one example, define

$$C_{\text{abs}} = \frac{\text{Im } S_1 - \text{Im } \Xi}{\text{Im } S_1} = \frac{\text{Im } \Delta \Xi}{\text{Im } S_1}$$

. The observable and sign convention must be fixed consistently in material-to-material comparisons.

For a one-insertion channel, suppose

$$K_{12}, K_{21} = O(\Gamma^{-2}), \quad S_1, S_2 = O(\Gamma^{-1})$$

and define

$$y = \left| \frac{\beta S_2}{\gamma_g} \right|$$

.

- $y \ll 1$:

$$\Delta \Xi = O(\Gamma^{-4}), \quad C_{\text{abs}} = O(\Gamma^{-3}).$$

- $y \gg 1$:

$$\Delta \Xi = O(\Gamma^{-3}), \quad C_{\text{abs}} = O(\Gamma^{-2}).$$

For fixed $\gamma_g \neq 0$ and fixed Ω_c , one ultimately has $y \rightarrow 0$, so the strict $\Gamma \rightarrow \infty$ limit is the former branch.

10. Full-Liouvillian EIT no-go and observable criterion

10.1 Pathway-resolved response operator

For a weak-probe problem with a unique steady state, define on the trace-zero response space

$$\mathbb{M} = -Q_0 \mathcal{L}_c Q_0$$

as the response generator, so that

$$\mathbb{M}|u^{(1)}\rangle\rangle = |b_p\rangle\rangle$$

. Let $\mathbb{S}$ be the long-lived coherence sector responsible for the target EIT mechanism, and let $\mathbb{X}$ be the complementary direct optical-response sector. Partition the response problem as

$$\mathbb{M} = \begin{pmatrix} A & B \\ C & G_g \end{pmatrix}, \quad |b_p\rangle\rangle = \begin{pmatrix} b_p \\ 0 \end{pmatrix}, \quad \langle\langle d_p| = (d^\dagger \quad 0)$$

.

Define the full response and the counterfactual response obtained by cutting only the targeted cross-feedback as

$$\chi_{\text{full}} = \chi_0 d^\dagger x, \quad \chi_{\text{cut}}^{(\mathbb{S})} = \chi_0 d^\dagger A^{-1} b_p$$

. The cut operation sets $B = C = 0$ while holding A, G_g, b_p, d and the zeroth-order state fixed. It is therefore generally different from switching the control field off.

10.2 Theorem 2B — Targeted full-Liouvillian sector-resolved EIT no-go

If A and

$$S_g = G_g - CA^{-1}B$$

are invertible at a regular point,

$$\delta\chi_{\mathbb{S}} \equiv \chi_{\text{full}} - \chi_{\text{cut}}^{(\mathbb{S})} = \chi_0 d^\dagger A^{-1} B S_g^{-1} C A^{-1} b_p$$

.

On a specified connected analytic parameter domain $\mathcal{U}$, if

$$\delta\chi_{\mathbb{S}}(\theta) \equiv 0 \quad (\theta \in \mathcal{U}_{\text{reg}})$$

then the net $\mathbb{S}$ -mediated contribution to the specified source-to-detector response vanishes, and the configuration is an **exact $\mathbb{S}$ -sector EIT no-go**. This does not require the internal block couplings B and C to vanish individually; complete cancellation at the detector is also included.

Conversely, $\delta\chi_{\mathbb{S}} \neq 0$ establishes permission for a net pathway through the specified sector, but it is not sufficient for a narrow transparency feature, positive contrast, or experimental visibility. Only when $\mathbb{S}$ is a long-lived coherence sector and B, C originate from the control Hamiltonian is this called coherent EIT-pathway permission.

For a closed scalar block with two lower states,

$$\delta\Xi_{\mathbb{S}} = \Xi_{\text{full}} - \Xi_{\text{cut}} = -\frac{\beta K_{12} K_{21}}{\gamma_g + \beta S_2}.$$

At a regular point with nonzero control and a finite denominator,

$$\delta\Xi_{\mathbb{S}} = 0 \iff K_{12} K_{21} = 0$$

.

Structural zeros versus pointwise zeros. A value $\delta\chi_{\mathbb{S}} = 0$ at one detuning or field value is not a domain-wide no-go. An exact no-go must be established as an identity using a reducing symmetry, finite Krylov orthogonality, an adjugate-numerator identity, or an analytic identity theorem.

The quantity $\delta\chi_{\mathbb{S}}$ is the difference between the full response and the $\mathbb{S}$ -cut response and is therefore a finite-dimensional rational transfer. When using a Krylov/moment theorem, apply it either to required cross-transfers such as $d^\dagger A^{-1} B$ and $CA^{-1} b_p$, or construct an augmented realization that represents the full/cut difference as a single transfer.

10.3 Sector dependence, invariance, and interpretation

10.3.1 Physical dependence on $\mathbb{S}$

The quantity $\delta\chi_{\mathbb{S}}$ depends on the specified projector pair

$$P_X + P_S = I, \quad P_X P_S = 0, \quad \text{Ran } P_S = \mathbb{S}$$

. In projector form,

$$\mathbb{M}_{\text{cut}}^{(\mathbb{S})} = P_X \mathbb{M} P_X + P_S \mathbb{M} P_S$$

and

$$\delta\chi_{\mathbb{S}} = \chi_0 \langle\langle d_p | \left(\mathbb{M}^{-1} - (\mathbb{M}_{\text{cut}}^{(\mathbb{S})})^{-1} \right) | b_p \rangle\rangle$$

. Therefore, every no-go statement is a statement about

$$(\mathfrak{C}, \mathbb{S}, \mathbb{X}, b_p, d_p)$$

and is not a system-wide scalar label independent of the sector choice.

10.3.2 Block-preserving basis invariance

Let $U = U_X \oplus U_S$ be an invertible basis transformation that does not mix $\mathbb{X}$ and $\mathbb{S}$. If the generator, source, and detector are transformed consistently,

$$\delta\chi'_{\mathbb{S}} = \delta\chi_{\mathbb{S}}$$

. Hence the result is independent of the basis chosen within either sector.

By contrast, if one performs a transformation that mixes $\mathbb{X}$ and $\mathbb{S}$ and then redefines the same coordinate cut, one has generally selected a different projector, and $\delta\chi$ changes. This is not basis dependence; it is a **change in the physical pathway definition**.

10.3.3 Algebraic counterfactual

The operator $\mathbb{M}_{\text{cut}}^{(\mathbb{S})}$ is an algebraic counterfactual used for pathway attribution and need not itself be an independent trace-preserving completely positive generator. Accordingly,

- do not identify $\chi_{\text{cut}}^{(\mathbb{S})}$ with the control-off spectrum;
- do not identify $\delta\chi_{\mathbb{S}}$ unconditionally with a directly measurable observable;
- for experimental comparison, use χ_{full} , control scans, two-photon dependence, ground-coherence dependence, and pole/residue analysis.

10.3.4 Additional conditions for a coherent interpretation

Generic blocks B, C can contain dissipative transfer as well as Hamiltonian coupling. The quantity $\delta\chi_{\mathbb{S}}$ is called a coherent EIT correction only if the following conditions hold.

1. $\mathbb{S}$ is a long-lived off-diagonal density-matrix sector between the selected lower states.
2. The EIT-relevant leading terms in B, C originate from the control Hamiltonian.
3. Population pumping and dissipative cross-transfer have been separated and audited in the full Liouvillian.

10.4 Total susceptibility zero is not an EIT no-go

For an ideal scalar Λ block, take

$$A = 1, \quad d_1 = d_2 = 1, \quad \beta = 1, \quad \gamma_g = 0$$

. Then

$$S_1 = S_2 = K_{12} = K_{21} = 1,$$

and hence

$$\Xi_{\text{cut}} = 1, \quad \Xi_{\text{full}} = 1 - \frac{1}{1} = 0.$$

On the other hand, define the transparency correction by

$$\Delta\Xi_{\text{tr}} \equiv \Xi_{\text{cut}} - \Xi_{\text{full}}$$

. Then

$$\Delta\Xi_{\text{tr}} = 1 \neq 0.$$

Therefore,

$$\boxed{\Xi_{\text{full}} = 0}$$

can represent perfect transparency. It must not be defined as a no-go condition.

10.5 General projector and singular formulation

If explicit (x, s) coordinates cannot be chosen, use response-space projectors P_X, P_S and define

$$\mathbb{M}_{\text{cut}}^{(\mathbb{S})} = P_X \mathbb{M} P_X + P_S \mathbb{M} P_S$$

. When the source and detector belong to the X sector,

$$\chi_{\text{full}} = \chi_0 \langle\langle d_p | \mathbb{M}^{-1} | b_p \rangle\rangle,$$

$$\chi_{\text{cut}}^{(\mathbb{S})} = \chi_0 \langle\langle d_p | (\mathbb{M}_{\text{cut}}^{(\mathbb{S})})^{-1} | b_p \rangle\rangle.$$

If zero modes or isolated singularities occur, do not use an ordinary inverse. Use a group/Drazin inverse on the compatible trace-zero subspace, or a Laurent finite part/removable continuation.

The partition $X \oplus S$ must be physically specified. Choosing a different coherence sector means classifying a different targeted EIT channel.

10.6 Observable EIT criterion

Even when the configuration is not an exact $\mathbb{S}$ -sector no-go, at least the following must be verified before claiming observable EIT.

1. $\delta\chi_{\mathbb{S}} \neq 0$ for the physically specified sector $\mathbb{S}$.
2. There is an absorption reduction, or a corresponding transmission increase, near two-photon resonance.
3. The feature depends on the ground-coherence decay rate.
4. Control-power scans, pole/residue analysis, linewidth scaling, or equivalent tests show that the feature cannot be explained solely by Autler–Townes splitting.
5. The signal exceeds the threshold after including ensemble averaging, propagation, background, and signal-to-noise ratio.

A control-on/off difference alone also contains population pumping, power broadening, and saturation. Use the pathway cut together with γ_g dependence, two-photon dependence, and pole tracking to isolate the coherent EIT component.

10.7 Validation tests

At minimum, compare the following on the same parameter grid.

1. A trace-zero constrained solve of the full GKSL Liouvillian and a group/Drazin solve
2. The full first-order Liouvillian response and the exact Schur-reduced formula
3. $\chi_{\text{full}} \chi_{\text{cut}}^{(\mathbb{S})} \delta\chi_{\mathbb{S}}$
4. In the textbook Λ limit,

$$\chi_{\text{full}} = 0, \quad \delta\chi_{\mathbb{S}} \neq 0$$

5. Agreement between the Mishina-type sum formula and direct block inversion for a diagonal multi-excited-state block
6. Validity of the S_g formulation even when G_g is singular
7. Invariance of $\delta\chi_{\mathbb{S}}$ under block-preserving basis transformations

8. The generic change in $\delta\chi$ when the projector is redefined after an X - S mixing transformation
The numerical tests in Section 21.3 are fixed as core solver regression tests.

11. Symmetry and group-theoretical criterion

11.1 Leading moment as projected dipole overlap

If $D \propto I$, or is scalar within the sector,

$$M_0 \propto p^\dagger c$$

. For a complete operative manifold E ,

$$p^\dagger c = \langle g_1 | \hat{d}_p^\dagger \Pi_E \hat{d}_c | g_2 \rangle.$$

Thus, the leading-order go/no-go result is determined by the lower-manifold matrix element of the effective tensor operator

$$\hat{O}_{pc} = \hat{d}_p^\dagger \Pi_E \hat{d}_c$$

.

11.2 Point-group selection rule

A nonzero matrix element requires

$$\Gamma(g_1)^* \otimes \Gamma(\hat{O}_{pc}) \otimes \Gamma(g_2)$$

to contain the totally symmetric representation. The representation $\Gamma(\hat{O}_{pc})$ is selected by the polarization tensor $\epsilon_p^* \otimes \epsilon_c$, the operative manifold, and any cavity projection.

11.3 Spin-scalar dipole

If the electric dipole operator is spin scalar to leading order,

$$\hat{\mathbf{d}} = \hat{\mathbf{d}}_{\text{orb}} \otimes I_{\text{spin}}$$

. For a lower-state pair that shares the same orbital state and differs only by orthogonal spin states, the scalar leading moment tends to vanish.

However, a finite higher-order moment can open in the presence of

- transverse spin-orbit coupling
- nonsecular spin-spin coupling
- hyperfine mixing
- transverse magnetic field
- spin-dependent electric-dipole correction
- incomplete manifold selection
- branch-dependent damping

Therefore, an exact no-go and asymptotic suppression must be distinguished in a physical system.

11.4 Caution concerning orbital degeneracy

An orbital-degenerate lower manifold is not sufficient for a go result. The following are required.

1. The lower-state symmetry permits the transition.
2. The polarization tensor contains the required irreducible representation.
3. The reduced dipole matrix element is nonzero.
4. The relevant spin overlap is nonzero.
5. The operative excited manifold has been selected correctly.
6. Damping and ground-state coherence permit practical visibility.

12. Classification taxonomy

Classify an arbitrary configuration $\mathfrak{C}$ as follows.

Class 0 — Unspecified / underdetermined

The required Hamiltonian, dipoles, jump operators, rates, or observable are insufficient, so no classification can be made.

Class I — Optical dark-subspace capable

$$\dim \ker \Omega > 0.$$

This establishes only the existence of an instantaneous optical dark superposition. A stationary dark state additionally requires the full Lindblad conditions of Section 3.4, and observable EIT is not guaranteed.

Class II — Exact sector-resolved EIT no-go

On the target domain,

$$\boxed{\delta\chi_{\mathbb{S}}(\theta) \equiv 0}$$

is proved by the finite Krylov criterion, an adjugate identity, or a reducing symmetry of the full generator that separates the targeted cross-transfer.

For a regular closed block with two lower states, this is equivalent to

$$K_{12}K_{21} \equiv 0$$

. A zero of the total susceptibility, $\chi_{\text{full}} = 0$, is not the definition of this class.

Class III — Asymptotic sector-mediated suppression

Let the first nonzero moments of the relevant cross-transfers be m_{12} and m_{21} , respectively, so that

$$K_{12} = O(\Gamma^{-m_{12}-1}), \quad K_{21} = O(\Gamma^{-m_{21}-1}).$$

At finite Γ the response is not exactly zero, but fast damping suppresses the targeted $\mathbb{S}$ -sector correction systematically. For example, in a scalar block with $S_2 = O(\Gamma^{-1})$,

$$\delta\Xi_{\mathbb{S}} = \begin{cases} O(\Gamma^{-m_{12}-m_{21}-2}), & |\beta S_2/\gamma_g| \ll 1, \\ O(\Gamma^{-m_{12}-m_{21}-1}), & |\beta S_2/\gamma_g| \gg 1. \end{cases}$$

When classifying only one transfer, record $K = O(\Gamma^{-m-1})$, where m is its first nonzero moment.

Class IV — Leading-order sector-kernel go

In the regular full-rank damping limit,

$$K_{12} = O(\Gamma^{-1}), \quad K_{21} = O(\Gamma^{-1})$$

and the product of the leading coefficients is nonzero, the targeted $\mathbb{S}$ -sector pathway is permitted at leading order. It is interpreted as a coherent pathway when $\mathbb{S}$ is a long-lived coherence and the cross-coupling is Hamiltonian in origin. This is still not sufficient for observable EIT.

Class V — Protected go

Under singular damping, if the leading protected-subspace transfer satisfies

$$\boxed{p^\dagger P(PA_0P)^{-1}Pc \neq 0}$$

then an $O(1)$ channel remains. Nonzero source and detector projections alone do not exclude phase cancellation.

Class VI — Practical no-go

$$K_{12}K_{21} \neq 0$$

Even if the kernel is nonzero, the full observable response may satisfy

$$|C_{\text{EIT}}| < C_{\min}$$

or fail the constraints imposed by signal-to-noise ratio, laser power, ionization, heating, or optical depth.

Class VII — Observable go

A narrow coherent transparency feature identifiable as EIT is obtained for the specified observable, normalization, and detection threshold.

Class VIII — Tunable transition

Fields, strain, polarization, cavities, or temperature move the configuration among Classes II–VII.

13. Standard Procedure for Applying the Theory to an Arbitrary Material

Step 1 — Fix the configuration

Specify the following uniquely.

- lower-state pair or lower manifold
- operative excited manifold
- probe/control frequencies and polarizations
- fieldstraincavity orientation
- temperature and bath condition
- measured observable
- targeted long-lived coherence sector $\mathbb{S}$

Step 2 — Construct the microscopic input

The required inputs are listed below.

category	required data
coherent structure	H_g, H_e fine structure Zeeman Stark strain hyperfine
optical coupling	complex dipole matrix elements selection rules polarization geometry
excited dissipation	radiative decay ISC pure dephasing orbital/spin relaxation
ground dissipation	T_1, T_2, T_2^* population exchange
disorder	strain distributions spectral diffusion orientation distribution
experimental	Ω_c optical depth sample length detection noise

Retain the complex phases, not only the absolute oscillator strengths.

Step 3 — Define the full master equation

$$\dot{\rho} = \mathcal{L}(\rho)$$

Construct the equation and distinguish, for every jump operator,

- jumps within a manifold
- loss out of the manifold
- correlated jump
- quasi-static disorder

.

Step 4 — Determine whether a reduced kernel is valid

If the fixed-ground optical-coherence block is invariant, use A_j^{-1} . Otherwise, use the full Liouville-space resolvent.

Step 5 — Compute the dark-state rank

$$\dim \mathcal{D}_{\text{dark}} = N_g - \text{rank } \Omega.$$

Step 6 — Compute the matrix kernel

$$\mathbf{K}(z, T, \mathbf{F}) = C^\dagger A^{-1} C$$

or its Liouville-space counterpart.

At minimum, save

$$S_1, S_2, K_{12}, K_{21}$$

on the same grid.

Step 7 — Test the targeted sector zero

Define the target sector $\mathbb{S}$, and construct $\chi_{\text{cut}}^{(\mathbb{S})}$ and $\delta\chi_{\mathbb{S}}$. Test an exact no-go in the following order.

1. A reducing symmetry that separates the targeted cross-transfer
2. finite Krylov moments
3. adjugate numerator identity
4. symbolic group-theory selection rule
5. A numerical zero scan only as supporting evidence

Do not use a zero of the total susceptibility as the no-go criterion.

Step 8 — Determine the strong-damping order

$$M_0, M_1, \dots$$

Compute the moments and identify the first nonzero one. For multiple rates, use the experimentally relevant trajectory $\Gamma_r(T)$.

Step 9 — Compute the full response and the sector-resolved correction

At minimum, compute

$$\chi_{\text{full}}, \quad \chi_{\text{cut}}^{(\mathbb{S})}, \quad \delta\chi_{\mathbb{S}}, \quad C_{\text{abs}}, \quad T$$

and convert them to the experimental observable, such as fluorescence, when necessary.

Step 10 — Distinguish EIT from ATS

Do not identify EIT from a transparency dip alone. At minimum, inspect

- ground-coherence dependence
- control-power dependence
- pole/eigenmode structure
- two-photon linewidth
- the difference from the pathway-cut response
- the difference from the control-off response

. Autler–Townes splitting generated by a strong control must be reported separately from coherent dark-state EIT.

Step 11 — Perform ensemble averaging correctly

For a quasi-static parameter ξ ,

$$\bar{\chi}(\omega) = \int d\xi P(\xi) \chi(\omega; \xi)$$

and average the observable response. A simple substitution of an average Hamiltonian or average linewidth can be invalid.

Step 12 — Report the classification and uncertainty

At minimum, output

$$\{\text{class}, m, K_{12}, K_{21}, \delta\chi_{\mathbb{S}}, C_{\text{EIT}}, \text{threshold margin}, \text{uncertainty}\}$$

.

14. Separation of the Markovian Main Scope, Static Broadening, and Out-of-Scope Extensions

14.1 Fast Markovian dynamics

When the system has the finite-dimensional time-independent GKSL generator and finite spectral gap assumed in the main theory, use resolvent, moment, and Schur-complement analyses.

14.2 Quasi-static disorder

Slow strain fluctuations, orientation distributions, and static spectral diffusion are not ergodic fast damping. Compute the response for each configuration and then perform the ensemble average.

Even for the same total linewidth,

- fast homogeneous exchange
- static inhomogeneous broadening

can exhibit different EIT survival.

14.3 Frequency-dependent self-energy — outside the main theorem

A genuinely non-Markovian bath lies outside the main theorem of this version. For formal comparison, write

$$A(z) = z - H_e - \Sigma(z)$$

and retain the transfer-kernel definition

$$K(z) = p^\dagger A(z)^{-1} c$$

.

However, to use the finite-moment theorem, one needs, for example, on the target band,

$$\Sigma(z) = \Gamma D + \Sigma_0(z)$$

and

$$\operatorname{Re} D \geq d_{\min} I > 0, \quad \|D^{-1} \Sigma_0(z)\| / \Gamma < 1$$

or another uniformly accretive factorization. In a Hermitian model, $D > 0$ is a special case.

Do not apply the finite-dimensional Cayley–Hamilton criterion without modification to an arbitrary non-Markovian bath, branch cut, or continuum.

14.4 Continuum excited manifolds — controlled approximations only

For a continuum or a very large vibronic manifold, use

- controlled truncation
- tensor-network/open-system solver
- spectral-density integral equation
- operator-valued transfer function

. Apply the finite-dimensional classification only as a controlled approximation after demonstrating convergence with respect to truncation.

15. Lifting a No-Go Condition by External Control

15.1 Generic perturbation

Add a sector-changing perturbation

$$V_{\text{mix}} = \lambda V_1 + O(\lambda^2)$$

to a system with an exact or leading-order no-go condition.

If one insertion opens a path between the source and readout sectors and the reciprocal kernels K_{12} and K_{21} open at the same order,

$$K_{12}, K_{21} = O(\lambda\Gamma^{-2}), \quad K_{12}K_{21} = O(\lambda^2\Gamma^{-4}).$$

For a non-normal or nonreciprocal model, determine the orders of the two kernels separately.

The canonical relative absorption contrast therefore scales as

$$C_{\text{abs}} = \begin{cases} O(\lambda^2\Gamma^{-3}), & y \ll 1, \\ O(\lambda^2\Gamma^{-2}), & y \gg 1. \end{cases}$$

15.2 Control knobs

Candidate controls for lifting a no-go condition or enhancing a go channel include

- transverse magnetic field
- spin-orbit mixing
- polarization rotation
- strain-induced eigenvector mixing
- cavity-selected submanifold
- spectral selection
- a Raman pathway through another manifold

15.3 General treatment of strain

Spin-scalar strain does not by itself generate a spin-changing first moment. It can nevertheless change

- detuning
- orbital eigenvectors
- spin-orbit admixture
- branch linewidth
- selected manifold

and therefore enhance or suppress an existing channel. No universal sign should be assigned to the effect of strain.

16. Status of Representative Channels in the Present Theory

16.1 NV^- zero-field ZPL spin- Λ

In a simplified parity- or sector-preserving model, a spin-scalar dipole and a common spin basis can produce an exact sector zero.

In the physical full 3E model, transverse spin-orbit coupling, nonsecular spin-spin coupling, hyperfine coupling, and related terms can break the sector. The defensible statement is therefore the asymptotic dissipative selection rule

$$M_0 = 0, \quad K = O(\Gamma^{-2})$$

. An all-orders exact no-go requires either a reducing symmetry of the full generator or the vanishing of every moment.

16.2 NV transverse-field channel

If a weak transverse field is parameterized by $\lambda \propto B_{\perp}$, the simplified model gives

$$K_{\text{even}} = O(B_{\perp} \Gamma^{-2}), \quad |K_{\text{even}}|^2 = O(B_{\perp}^2 \Gamma^{-4})$$

. Even when an odd channel vanishes in a parity-preserving model, the residual generated by realistic perturbations must be evaluated separately.

The numerical result currently obtained in the simplified model is

- $M_0 = 0$
- even channel: $M_1 \simeq 139 \text{ MHz}$
- odd channel: $M_1 \simeq 0$
- $|K| \propto B_{\perp}$
- $K \propto \Gamma^{-2}$

. This is a model-specific result that has not yet been verified with the full microscopic NV Hamiltonian and is therefore not part of the universal theorem.

16.3 Same-spin orbital- Λ

Within the same internal-spin sector, if

$$d_1^{\dagger} D^{-1} d_2 \neq 0$$

then

$$K_{12} = O(\Gamma^{-1})$$

and the channel is a leading-order kernel go. Candidate platforms such as group-IV centers, orbital doublets, and rare-earth crystal-field states require their own dipole tensors, phonon relaxation, and ground-coherence parameters before a practical-go classification can be made.

16.4 Atomicrare-earthmolecularquantum-dot systems

The same framework can be used, although the dominant physics differs.

- atomic vapor: Doppler averagecollisionsZeeman manifold
- rare-earth ion: crystal-field/hyperfine statesinhomogeneous broadeninglong optical lifetime
- molecule: vibronic continuumFranck–Condon phasesnon-Markovian bath
- quantum dot: fine-structure splittingcharge noisecavity and polarization selection

These systems are classified by configuration tuple rather than by material name.

17. Algorithmic decision tree

The following is the standard classification algorithm.

INPUT:

H_g, H_e , dipole vectors C , jump operators $\{L_{\mu}\}$,
field/strain/polarization, temperature-dependent rates,
probe/control parameters, observable definition

1. Build the full zeroth-order Liouvillian L_c and solve $\rho^{\wedge}(0)$.
2. Restrict the first-order problem to the compatible trace-zero response space.
3. Define the targeted long-lived coherence sector S and complementary response sector X .
4. Test whether fixed-ground optical-coherence blocks are invariant.
 - YES: construct reduced $A_j(z)$.

- NO: retain the full Liouville-space block.
- 5. Compute optical dark-state dimension: $N_g - \text{rank}(\Omega)$.
- 6. Compute S_i and K_{ij} where the reduced kernel is valid.
- 7. Construct χ_{full} , pathway-cut χ_{cut} , and $\delta\chi_S$.
- 8. Test exact targeted zero:
 - a. reducing symmetry of the targeted cross-transfer;
 - b. finite Krylov moments;
 - c. adjugate numerator identity;
 - d. Laurent/Drazin test at singular points.
- 9. If not exact zero, decompose $A = \Gamma D + A_0$.
- 10. Determine whether D is full rank.
 - full rank: compute moment hierarchy and suppression order.
 - singular: compute protected-subspace Schur complement.
- 11. For multiple rates, follow the physical trajectory $\Gamma_r(T)$.
- 12. Compute observable absorption/transmission/fluorescence.
- 13. Distinguish EIT from ATS and population/saturation effects.
- 14. Ensemble-average static disorder at the response level.
- 15. Compare contrast/SNR with the experimental threshold.

OUTPUT:

optical dark-state rank,
 exact/asymptotic targeted-kernel class and order,
 K_{12} , K_{21} , S_1 , S_2 ,
 χ_{full} , χ_{cut} , $\delta\chi_S$,
 observable EIT contrast,
 EIT/ATS label,
 parameter margin and uncertainty,
 exact assumptions and projectors used.

A zero of the total response is not used as a no-go criterion. A no-go is identified as an identically vanishing targeted coherence feedback.

18. Final theorem package — corrected finite-dimensional version

Theorem 1A — Optical dark-subspace rank

$$\mathcal{D}_{\text{opt}} = \ker \Omega, \quad \dim \mathcal{D}_{\text{opt}} = N_g - \text{rank } \Omega.$$

This theorem concerns instantaneous optical decoupling and does not by itself guarantee stationary EIT.

Theorem 1B — Stationary pure Lindblad dark state

A pure state $|D\rangle\langle D|$ is stationary if and only if it is a common eigenvector of every jump operator and an eigenvector of the shifted Hamiltonian. An EIT dark state additionally requires $\Omega|D\rangle = 0$.

Theorem 2A — Full-to-reduced weak-probe susceptibility

The full linear response is

$$\rho^{(1)} = -\mathcal{L}_c^\# \mathcal{V}_p \rho^{(0)}.$$

. For a closed two-lower-state block, the Schur complement gives

$$\Xi = S_1 - \frac{\beta K_{12} K_{21}}{\gamma_g + \beta S_2}.$$

Theorem 2B — Targeted full-Liouvillian sector-resolved EIT no-go

Write the first-order response block as

$$\mathbb{M} = \begin{pmatrix} A & B \\ C & G_g \end{pmatrix}, \quad S_g = G_g - CA^{-1}B$$

. At a regular point,

$$\delta\chi_{\mathbb{S}} = \chi_{\text{full}} - \chi_{\text{cut}}^{(\mathbb{S})} = \chi_0 d^\dagger A^{-1} B S_g^{-1} C A^{-1} b_p.$$

If $\delta\chi_{\mathbb{S}} \equiv 0$ on the specified domain, the net source-to-detector contribution of the sector $\mathbb{S}$ vanishes and the configuration is an exact $\mathbb{S}$ -sector EIT no-go. The condition $\chi_{\text{full}} = 0$ is not a no-go condition and may represent perfect EIT.

Theorem 3 — Dissipative moment hierarchy

If $A_\Gamma = \Gamma D + A_0$, D is invertible, and the large-rate expansion converges, then a first nonzero moment M_m gives

$$K_\Gamma = M_m \Gamma^{-m-1} + O(\Gamma^{-m-2}).$$

Theorem 4 — Exact finite-dimensional transfer zero

For response dimension N ,

$$M_0 = \cdots = M_{N-1} = 0$$

is equivalent to finite Krylov orthogonality and to

$$K_\Gamma = 0 \quad (\Gamma \notin \text{spec } X)$$

. This conclusion is not restricted to the Neumann-series convergence domain. For an EIT no-go application, choose K_Γ to be the targeted sector cross-transfer.

Theorem 5 — Full-generator reducing-symmetry transfer zero

If the full Liouvillian or reduced generator has orthogonal reducing sectors and the source and detector lie in different sectors, the group-inverse/resolvent transfer vanishes exactly. To obtain an EIT no-go, define the source and detector for the targeted coherence-feedback transfer.

Theorem 6 — Singular-damping protected channel

Under $D = D^\dagger \geq 0$, the Schur complement on $\ker D$ determines the leading response.

$$K_\Gamma = p^\dagger P(PA_0P)^{-1}Pc + O(\Gamma^{-1}).$$

If the leading scalar is nonzero, an $O(1)$ protected channel remains.

Theorem 7 — Adjugate, analytic identity, and singular transfer criterion

On the regular set of a connected holomorphic or real-analytic domain,

$$p^\dagger \text{adj } A(\theta)c \equiv 0$$

gives an exact transfer zero. At a singular point, distinguish Laurent pole coefficients, finite parts, removable continuations, and the Drazin inverse; do not interpret $0/0$ as zero.

Corollary — Symmetric two-branch hopping-rate map

$$(p_X - p_Y) = -\Gamma_{XY}(p_X - p_Y)$$

For the symmetric hopping model defined by

$$\boxed{\gamma_{oc} = \Gamma_{XY}/4}.$$

Corollary — Weak sector-breaking control

the additional damping of each optical coherence is

$$K_{12}, K_{21} = O(\lambda\Gamma^{-2})$$

If reciprocal kernels open at the same order through one sector-changing insertion,

$$O(\lambda^2\Gamma^{-3}) \quad \text{or} \quad O(\lambda^2\Gamma^{-2})$$

. In a nonreciprocal model, evaluate K_{12} and K_{21} separately.

19. Falsifiable predictions and standard outputs

To make the theory experimentally falsifiable, output the following quantities for every material.

19.1 Thermal/damping collapse

$$K_{12}(T), \quad K_{21}(T), \quad \delta\chi_S(T), \quad C_{\text{EIT}}(T)$$

as functions of the material-specific fast-rate ratio

$$x(T) = \Gamma_{\text{fast}}(T)/\Delta_{\text{sel}}$$

. Use a single universal crossover function only within the model class for which it has been derived.

19.2 External-field opening

At small field, measure scaling laws such as

$$|K|^2 \propto B_{\perp}^2$$

and the exponent associated with the relevant visibility branch.

19.3 Polarization null test

Compare a polarization for which symmetry enforces a zero with one for which the response is nonzero.

19.4 Fast exchange vs static broadening

Test whether two conditions with the same observed linewidth exhibit different EIT survival.

19.5 Material comparison table

For each configuration, construct

configuration	dark rank	first nonzero moment	targeted sector class	predicted contrast	evidence status
material/channel A			exact/asymptotic/go		measured/predicted

.

20. Current Theoretical Limitations

The following lie outside the integrated theory or require additional proof.

1. A finite-moment exact theorem for an arbitrary infinite-dimensional non-Markovian bath
2. A simple kernel formula under arbitrary strong-probe saturation
3. Collective EIT for which propagation, cavity feedback, or many-body interactions cannot be neglected
4. Inferring an all-orders exact zero from a numerical zero
5. Inferring observable EIT directly from $K \neq 0$
6. Classifying EIT, CPT, STIRAP, Raman transfer, and quantum memory under one identical condition
7. Assigning one go/no-go label to an entire material family
8. Defining a unique full-Liouvillian EIT no-go without specifying the coherence-sector partition
9. Interpreting a zero of the total susceptibility as an EIT no-go

STIRAP requires not only a nonzero Raman kernel but also a time-dependent dark eigenstate, a counterintuitive pulse sequence, and an adiabatic gap.

21. Theory-completion audit and remaining validation tasks

21.1 Completion status

The term “completed” in this version refers only to the stated assumptions of finite dimensionality, Markovian GKSL dynamics, a stationary rotating frame, weak probing, and an analytic parameter domain.

requested item	status in v6.2	exact scope
Dark-state theorem	completed	Separates optical nullity from a stationary Lindblad dark state and gives necessary and sufficient conditions
Exact finite-dimensional transfer-zero proof	completed	Extended to every invertible Γ using Cayley–Hamilton, Krylov, and adjugate arguments
Full-Liouvillian EIT no-go definition	completed	Defined as the sector-resolved condition $\delta\chi_S \equiv 0$, not as a zero of the total susceptibility
Adjugate singularities	completed	Separates the regular domain, Laurent poles, finite parts, removable continuations, and Drazin treatment
Weak-probe susceptibility	completed	Derived from the full-Liouvillian group inverse and proved in scalar/matrix Schur-complement form
Singular ground block	completed	Uses $S_g = G_g - CA^{-1}B$ without requiring G_g^{-1}

requested item	status in v6.2	exact scope
Γ_{XY} mapping	completed for the stated convention	For a symmetric population-imbalance rate, $\gamma_{oc} = \Gamma_{XY}/4$
Scope of $D > 0$	completed	Separates invertibility, strict accretivity, and the Hermitian singular theorem
Full-generator symmetry	completed as a transfer theorem	Audits every jump operator, the steady-state projector, source, and detector, not only the Hamiltonian
Analytic identity condition	completed	Restricted to holomorphic or real-analytic domains, without overextending to smooth dependence
Sector dependence	completed	States the $\mathbb{S}$ dependence, block-preserving basis invariance, and dependence under sector mixing
Textbook Λ recovery	completed	Recovers the standard three-level susceptibility exactly
Mishina-type recovery	completed	Recovers the diagonal multi-excited resolvent sum and inhomogeneous averaging
Full/reduced unit test	completed	Compares a physical three-level GKSL Liouvillian with the Schur formula to machine precision

21.2 Theoretical conditions for proceeding to numerical calculations

The present version satisfies the following conditions.

1. Optical dark states and stationary dark states are not conflated.
2. An exact transfer zero does not depend on the convergence domain of the large-rate series.
3. An ordinary inverse is not used for a singular Liouvillian.
4. The reduced susceptibility is identified as the Schur complement of the full first-order response.
5. A Schur complement valid for singular G_g is used.
6. The coefficient relating a population-hopping rate to an optical-coherence damping rate is derived from the jump convention.
7. The use of positivity assumptions is restricted to the locations where they are required.
8. The symmetry theorem is formulated as a reducing-sector transfer theorem for the full generator.
9. A zero of the total response is separated from an EIT no-go.
10. The targeted sector $\mathbb{S}$, $\chi_{\text{cut}}^{(\mathbb{S})}$, and $\delta\chi_{\mathbb{S}}$ are explicitly specified.
11. The sector dependence and block-preserving basis invariance of $\delta\chi_{\mathbb{S}}$ are stated explicitly.
12. The textbook Λ model and a Mishina-type diagonal multilevel response are recovered as special cases.
13. Unit tests comparing the full GKSL Liouvillian and reduced formula have been completed to machine precision.

Accordingly, there is no need to prioritize further abstract generalization before proceeding to material-specific numerical calculations.

21.3 Completed full-Liouvillian and reduced-formula unit tests

The core regression script `eit_unit_tests_v6_2.py` independently verifies the following results using double-precision complex arithmetic.

Test U1 — Physical three-level GKSL Liouvillian vs reduced Λ formula

Using the basis $(|g_1\rangle, |g_2\rangle, |e\rangle)$,

$$\gamma_{31} = 0.73, \quad \gamma_{21} = 0.041, \quad \Delta_p = 0.37, \quad \delta_2 = -0.083, \quad \Omega_c = 1.16$$

was adopted. A full GKSL Liouvillian containing excited-state decay jumps and a $|g_2\rangle \rightarrow |g_1\rangle$ relaxation jump was constructed, and the zeroth-order steady state and trace-zero first-order response were solved directly.

$$\rho_{eg_1, \text{full}}^{(1)} = -0.09734305655387952 + 0.06276778568576767i,$$

$$\rho_{eg_1, \text{reduced}}^{(1)} = -0.09734305655387981 + 0.06276778568576774i.$$

$$\boxed{\varepsilon_{\text{rel}} = 2.996 \times 10^{-16}}$$

Thus, the full-Liouvillian weak-probe solution and the reduced formula of Section 9.6 agree to machine precision.

Test U2 — Textbook perfect-EIT limit

$$\gamma_{21} = \delta_2 = 0, \quad \beta > 0$$

one obtains

$$\Xi_{\text{full}} = 0, \quad \Xi_{\text{cut}} = A^{-1}, \quad \delta\Xi_{\mathbb{S}} = -A^{-1} \neq 0.$$

For the numerical example $A = 0.9 - 0.2i$,

$$\Xi_{\text{cut}} = 1.0588235294 + 0.2352941176i.$$

This fixes the convention so that a zero of the total susceptibility is not misidentified as a no-go.

Test U3 — Mishina-type diagonal multi-excited-state recovery

For a random complex diagonal generator with four excited branches, the kernel sum of Section 9.7 was compared with direct inversion of the $(N_e + 1)$ -dimensional block.

$$\Xi_{\text{kernel}} = 1.9240573122129712 - 0.06390491266120876i,$$

$$\Xi_{\text{direct}} = 1.9240573122129716 - 0.06390491266120868i,$$

$$\boxed{\varepsilon_{\text{rel}} = 2.347 \times 10^{-16}}.$$

Test U4 — Block-preserving basis invariance

For a random $3 \oplus 2$ block system, independent unitaries $U_X \oplus U_S$ were applied.

$$\boxed{\frac{|\delta\chi'_{\mathbb{S}} - \delta\chi_{\mathbb{S}}|}{\max(1, |\delta\chi_{\mathbb{S}}|)} = 2.734 \times 10^{-16}}.$$

Test U5 — Sector-mixing dependence

After a unitary transformation mixing X and S , the coordinate cut was redefined, yielding

$$\delta\chi_{\text{before}} = -0.0049367 + 0.1224286i,$$

$$\delta\chi_{\text{after recut}} = -0.2348257 - 0.0281825i,$$

$$|\Delta| = 0.2748321.$$

This numerically confirms that $\delta\chi_{\text{S}}$ is invariant under basis changes within the sectors but depends on the physical definition of the sector.

Existing algebraic regression tests

test	result
exact Krylov transfer zero	residual 0
one-insertion hierarchy	fitted slope -2.0000000000000004
symmetric hopping map	$\gamma_{\text{oc}} - \Gamma_{XY}/4 = 0$
scalar weak-probe Schur complement	relative residual 1.62×10^{-16}
matrix Schur complement with singular G_g	residual 0
Drazin finite part	regularization error 2.50×10^{-4} at finite shift
symmetry-projector transfer zero	residual 0

These tests are consistency checks of the finite-dimensional theorems and solver architecture; they do not replace validation of the material-specific Hamiltonian, jump set, rate convention, or observable model.

21.4 Remaining material-specific numerical and model validation

Priority A — Material-solver regression and observable validation

The core full/reduced unit tests are complete. When moving to a material model, maintain the following as regression tests for every parameter set.

- Domain of agreement between the full-Liouvillian group/Drazin solve and the reduced block
- Reconstruction error among χ_{full} , $\chi_{\text{cut}}^{(\text{S})}$, and $\delta\chi_{\text{S}}$
- Krylov exact-zero and Laurent/singular-point tests
- Nonnormal resolvent norm and condition number
- EIT/ATS pole-residue classifier
- Audit of truncation dimension, steady-state uniqueness, and trace preservation

Priority B — NV microscopic audit

- full six-level 3E Hamiltonian
- physical complex dipole vectors
- Mapping of $k_{X \leftrightarrow Y}(T)$ from the jump convention
- full-generator symmetry audit table
- Verification of M_0, M_1 and the predicted suppression order
- transverse-fieldstrain symmetry breaking
- Comparison of $\chi_{\text{full}}, \chi_{\text{cut}}^{(\text{S})}, \delta\chi_{\text{S}}$

Priority C — Cross-material validation

Using the same solver and observable normalization, demonstrate

1. exact targeted no-go case
2. asymptotic no-go case
3. a leading-order or protected go case

4. experimentally established EIT benchmark
5. field/polarization tunable escape case

Assessment: Within the stated assumptions, the principal mathematical inconsistencies in the general theory have been corrected. The main remaining work is not the addition of further abstract theorems, but the validation of microscopic inputs and solvers and the production of material-specific numerical predictions.

22. Recommended Central Claim of the Paper

English

For any finite-dimensional Markovian open quantum emitter admitting a stationary rotating-frame weak-probe response, and for a physically specified long-lived sector S , the S -mediated probe-response modification is a finite-dimensional transfer function. Exact S -sector no-go corresponds to an identically vanishing sector-resolved correction, not to a vanishing total susceptibility. The framework recovers the textbook three-level Λ susceptibility and the diagonal multi-excited-state Mishina-type response. Fast full-rank damping produces a hierarchy whose suppression order is fixed by the first nonvanishing resolvent moment, while observable EIT requires the full Liouvillian visibility budget.

Japanese version

For a finite-dimensional Markovian open quantum emitter admitting a weak-probe response in a stationary rotating frame, the probe-response modification mediated by a physically specified long-lived sector S is a finite-dimensional transfer function. If this sector-resolved correction vanishes identically, the system is an exact S -sector no-go; a zero of the total susceptibility is not a no-go condition. The framework recovers the textbook three-level Λ susceptibility and a Mishina-type diagonal multi-excited-state response as special cases. Under strong full-rank damping, the first nonzero resolvent moment determines the suppression order, while observable EIT requires a separate visibility condition based on the full Liouvillian.

23. Final Conclusion

The theory has been extended from an NV-specific no-go argument to a material-independent framework containing the following elements.

1. Hilbert-space dark-state rank
2. basis-invariant matrix resolvent kernel
3. full Liouville-space pathway-resolved response correction
4. a moment hierarchy for full-rank damping
5. exact Krylov/adjugate/symmetry transfer-zero criteria
6. protected channels under singular damping
7. multiple-rate and external-field phase diagram
8. observable-specific EIT visibility
9. separation of static broadening from fast dynamics
10. a channel-resolved taxonomy

The theory can therefore be applied to arbitrary materials. The final classification, however, is performed not by material name but for each configuration after fixing the material, lower-state pair, excited manifold, polarizations, external fields, temperature, observable, and targeted coherence sector. An exact EIT no-go is $\delta\chi_S \equiv 0$, not $\chi_{\text{full}} = 0$. This is the strongest generalization that does not overstate the theory.

Appendix A. Notation

symbol	meaning
$\mathcal{H}_g, \mathcal{H}_e$	lower and excited manifolds
Ω	instantaneous optical coupling map
C	matrix of dipole coupling vectors
$A(z)$	optical-coherence generator
$G = A^{-1}$	resolvent/Green operator
K_{ab}	off-diagonal coherent-response kernel
S_a	diagonal optical response
D	fast damping-shape operator
Γ	fast optical-coherence damping scale
$M_n = p^\dagger X^n \nu$	resolvent moment
Q	reducing symmetry operator
$P = \text{Proj}(\ker D)$	protected-subspace projector
γ_g	complex lower-coherence decay/detuning
$\beta = \Omega_c ^2/4$	control intensity parameter
Ξ	normalized full local probe response
$\chi_{\text{cut}}^{(\mathbb{S})}$	algebraic counterfactual response obtained by cutting the cross-feedback with the specified sector
$\delta\chi_{\mathbb{S}}$	$\chi_{\text{full}} - \chi_{\text{cut}}^{(\mathbb{S})}$, sector-mediated response modification
C_{abs}	specified relative absorption contrast

Appendix B. Proof-repair ledger

The principal points corrected and fixed in this version are as follows.

- The dark-state rank is restricted to an optical-nullity theorem, while the full Lindblad necessary and sufficient conditions for a stationary pure state are retained.
- The moment symbol is standardized as $\nu = D^{-1}c$.
- An exact transfer zero is proved through the Krylov/Cayley–Hamilton equivalence.
- The adjugate criterion is restricted to an analytic regular domain.
- At singular points, finite parts, removable continuations, and ordinary inverses are distinguished.
- The weak-probe formula is derived from the full-Liouvillian group-inverse response.
- The formulation is corrected to use $S_g = G_g - CA^{-1}B$, without requiring the inverse of the scalar ground block.
- The full-Liouvillian EIT no-go is corrected from $\chi_{\text{full}} = 0$ to $\delta\chi_{\mathbb{S}} \equiv 0$.
- The ideal Λ system is added as a counterexample, confirming that a zero total response may represent perfect EIT.
- The $\Gamma_{XY}/4$ mapping is verified explicitly for symmetric two-branch hopping.
- The locations requiring $D > 0$ are separated from those requiring only invertibility or strict accretivity.
- The symmetry theorem is positioned as a generic transfer-zero theorem, and its use for an EIT no-go is restricted to the targeted cross-transfer.
- Reciprocal and nonreciprocal kernels are distinguished in weak sector-breaking scaling.

Appendix C. Provenance and integration status

This integrated version is based on the v6.0 theory package and incorporates into the v6.1 proof-corrected formulation the sector dependence, recovery of standard models, Markovian scope, and full/reduced unit tests.

The main integrated elements are as follows.

- Excited-confined jumps in the optical-coherence block are treated as full-rank damping.
- Exact dark states, coherent pathways, and observable EIT are separated.
- Exact transfer zeros are classified by finite moments, Krylov orthogonality, adjugate identities, and reducing symmetries.
- An EIT no-go is defined as an identically vanishing pathway-cut correction $\delta\chi_{\mathbb{S}}$ for a physically specified sector $\mathbb{S}$.
- A zero of the total susceptibility is excluded from the no-go condition.
- A singular ground-coherence block is handled by the A -side Schur complement.

- Multiple rates, singular damping, and static disorder are treated within the main scope, while non-Markovian extensions are stated explicitly to be outside it.
- No-go/go classifications are made at the configuration/channel level rather than the material level.

Integration status: Within the stated finite-dimensional Markovian stationary weak-probe assumptions, the theory has been corrected into an integrated form suitable for numerical calculations. Material-specific conclusions must be finalized only after numerical validation of each Hamiltonian, dipole operator, jump operator, rate, and observable model.